Healthcare Project

May 8, 2026

1 1.Import Libaries that i am going to use

[ ]:
```
import pandas as pd
import numpy as np
import seaborn as sns
```

1.0.1 2.Reading Healthcare admission Dataset as an Excel file

[4]:
```
df = pd.read_excel(r"C:\Users\Nkhosikhona Mpungose\OneDrive - Insight␣
↪forge\Desktop\Data Analysis\Excel Course\Healthcare_Department_Data.xlsx")
df
```

[4]:
```
   Patient_ID        Department  Visit_Date Doctor_Name         Diagnosis  \
0        P001  General Medicine  2025-01-01  Dr. Mthembu               Flu
1        P002        Pediatrics  2025-01-02   Dr. Naidoo          Check-up
2        P003         Emergency  2025-01-03  Dr. Khumalo          Fracture
3        P004           Surgery  2025-01-04  Dr. Daniels  Surgery Follow-up
4        P005          Pharmacy  2025-01-05   Dr. Pillay      Prescription
5        P006  General Medicine  2025-01-06  Dr. Mthembu               Flu
6        P007        Pediatrics  2025-01-07   Dr. Naidoo          Check-up
7        P008         Emergency  2025-01-08  Dr. Khumalo          Fracture
8        P009           Surgery  2025-01-09  Dr. Daniels  Surgery Follow-up
9        P010          Pharmacy  2025-01-10   Dr. Pillay      Prescription
10       P011  General Medicine  2025-01-11  Dr. Mthembu               Flu
11       P012        Pediatrics  2025-01-12   Dr. Naidoo          Check-up
12       P013         Emergency  2025-01-13  Dr. Khumalo          Fracture
13       P014           Surgery  2025-01-14  Dr. Daniels  Surgery Follow-up
14       P015          Pharmacy  2025-01-15   Dr. Pillay      Prescription
15       P016  General Medicine  2025-01-16  Dr. Mthembu               Flu
16       P017        Pediatrics  2025-01-17   Dr. Naidoo          Check-up
17       P018         Emergency  2025-01-18  Dr. Khumalo          Fracture
18       P019           Surgery  2025-01-19  Dr. Daniels  Surgery Follow-up
19       P020          Pharmacy  2025-01-20   Dr. Pillay      Prescription
20       P021  General Medicine  2025-01-21  Dr. Mthembu               Flu
21       P022        Pediatrics  2025-01-22   Dr. Naidoo          Check-up
22       P023         Emergency  2025-01-23  Dr. Khumalo          Fracture
23       P024           Surgery  2025-01-24  Dr. Daniels  Surgery Follow-up
```

24	P025	Pharmacy	2025-01-25	Dr. Pillay	Prescription
25	P026	General Medicine	2025-01-26	Dr. Mthembu	Flu
26	P027	Pediatrics	2025-01-27	Dr. Naidoo	Check-up
27	P028	Emergency	2025-01-28	Dr. Khumalo	Fracture
28	P029	Surgery	2025-01-29	Dr. Daniels	Surgery Follow-up
29	P030	Pharmacy	2025-01-30	Dr. Pillay	Prescription
30	P031	General Medicine	2025-01-31	Dr. Mthembu	Flu
31	P032	Pediatrics	2025-02-01	Dr. Naidoo	Check-up
32	P033	Emergency	2025-02-02	Dr. Khumalo	Fracture
33	P034	Surgery	2025-02-03	Dr. Daniels	Surgery Follow-up
34	P035	Pharmacy	2025-02-04	Dr. Pillay	Prescription
35	P036	General Medicine	2025-02-05	Dr. Mthembu	Flu
36	P037	Pediatrics	2025-02-06	Dr. Naidoo	Check-up
37	P038	Emergency	2025-02-07	Dr. Khumalo	Fracture
38	P039	Surgery	2025-02-08	Dr. Daniels	Surgery Follow-up
39	P040	Pharmacy	2025-02-09	Dr. Pillay	Prescription
40	P041	General Medicine	2025-02-10	Dr. Mthembu	Flu
41	P042	Pediatrics	2025-02-11	Dr. Naidoo	Check-up
42	P043	Emergency	2025-02-12	Dr. Khumalo	Fracture
43	P044	Surgery	2025-02-13	Dr. Daniels	Surgery Follow-up
44	P045	Pharmacy	2025-02-14	Dr. Pillay	Prescription
45	P046	General Medicine	2025-02-15	Dr. Mthembu	Flu
46	P047	Pediatrics	2025-02-16	Dr. Naidoo	Check-up
47	P048	Emergency	2025-02-17	Dr. Khumalo	Fracture
48	P049	Surgery	2025-02-18	Dr. Daniels	Surgery Follow-up
49	P050	Pharmacy	2025-02-19	Dr. Pillay	Prescription

	Treatment_Cost	Patient_Satisfaction_Score
0	350	4
1	200	5
2	2200	3
3	5000	4
4	120	5
5	350	4
6	200	5
7	2200	3
8	5000	4
9	120	5
10	350	4
11	200	5
12	2200	3
13	5000	4
14	120	5
15	350	4
16	200	5
17	2200	3
18	5000	4

19	120	5
20	350	4
21	200	5
22	2200	3
23	5000	4
24	120	5
25	350	4
26	200	5
27	2200	3
28	5000	4
29	120	5
30	350	4
31	200	5
32	2200	3
33	5000	4
34	120	5
35	350	4
36	200	5
37	2200	3
38	5000	4
39	120	5
40	350	4
41	200	5
42	2200	3
43	5000	4
44	120	5
45	350	4
46	200	5
47	2200	3
48	5000	4
49	120	5

2 3.Data Information

[8]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 7 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Patient_ID                  50 non-null     str
 1   Department                  50 non-null     str
 2   Visit_Date                  50 non-null     str
 3   Doctor_Name                 50 non-null     str
 4   Diagnosis                   50 non-null     str
```

```
 5   Treatment_Cost              50 non-null     int64
 6   Patient_Satisfaction_Score  50 non-null     int64
dtypes: int64(2), str(5)
memory usage: 2.9 KB
```

2.0.1 4.Count of Duplicates

[11]:
```
df.duplicated().sum()
```

[11]:
```
np.int64(0)
```

2.0.2 5.View the Columns

[15]:
```
df.columns
```

[15]:
```
Index(['Patient_ID', 'Department', 'Visit_Date', 'Doctor_Name', 'Diagnosis',
       'Treatment_Cost', 'Patient_Satisfaction_Score'],
      dtype='str')
```

3 6.filter columns Diagnosis & Treatment cost

[19]:
```
df2 = df.filter(['Diagnosis',
       'Treatment_Cost'])
df2
```

[19]:

	Diagnosis	Treatment_Cost
0	Flu	350
1	Check-up	200
2	Fracture	2200
3	Surgery Follow-up	5000
4	Prescription	120
5	Flu	350
6	Check-up	200
7	Fracture	2200
8	Surgery Follow-up	5000
9	Prescription	120
10	Flu	350
11	Check-up	200
12	Fracture	2200
13	Surgery Follow-up	5000
14	Prescription	120
15	Flu	350
16	Check-up	200
17	Fracture	2200
18	Surgery Follow-up	5000
19	Prescription	120

```
20                Flu             350
21           Check-up             200
22           Fracture            2200
23  Surgery Follow-up            5000
24       Prescription             120
25                Flu             350
26           Check-up             200
27           Fracture            2200
28  Surgery Follow-up            5000
29       Prescription             120
30                Flu             350
31           Check-up             200
32           Fracture            2200
33  Surgery Follow-up            5000
34       Prescription             120
35                Flu             350
36           Check-up             200
37           Fracture            2200
38  Surgery Follow-up            5000
39       Prescription             120
40                Flu             350
41           Check-up             200
42           Fracture            2200
43  Surgery Follow-up            5000
44       Prescription             120
45                Flu             350
46           Check-up             200
47           Fracture            2200
48  Surgery Follow-up            5000
49       Prescription             120
```

4 7.Group Treatment Costs by Diagnosis

[53]:
```
df4=df2.groupby('Diagnosis').sum()
df4
```

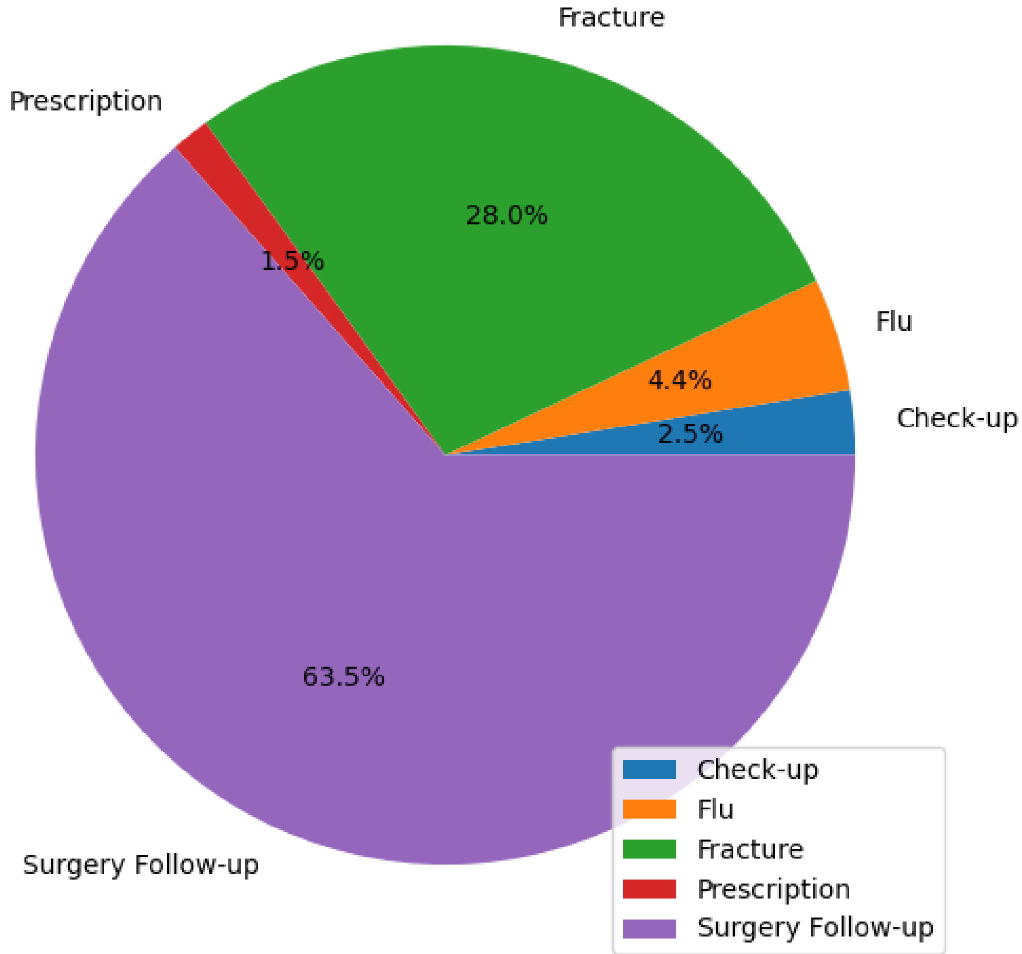
[53]:
```
                   Treatment_Cost
Diagnosis
Check-up                     2000
Flu                          3500
Fracture                    22000
Prescription                 1200
Surgery Follow-up           50000
```

5 Distribution of Treatment Costs by Diagnosis

Surgery Follow-up accounts for the largest share of total treatment costs, contributing approximately 63.5% of overall expenditure. This indicates that post-surgical care is the most cost-intensive service, likely due to specialized medical attention, extended recovery support, and follow-up procedures. Fracture treatments represent the second-highest cost, making up about 28.0% of total treatment expenses. This suggests that injury-related cases require significant resources, possibly including imaging, orthopedic procedures, and rehabilitation services. In contrast, Flu treatments and Check-ups contribute relatively small portions of the total cost, accounting for roughly 4.4% and 2.5% respectively. These conditions are generally routine and less resource-intensive, resulting in lower overall costs despite potentially higher patient volumes. Prescriptions make up the smallest share at approximately 1.5%, indicating minimal cost impact compared to other diagnoses. Overall, the chart clearly shows that complex and specialized treatments drive the majority of healthcare costs, while routine services contribute only marginally. This insight can help hospital management focus cost-control strategies on high-expense areas such as surgical follow-ups and fracture care, while maintaining efficiency in lower-cost services.

[57]:
```
df4.plot.pie(x = 'Diagnosis', y= 'Treatment_Cost', figsize = [7,7],autopct='%1.
↪1f%%')
```

[57]:
```
<Axes: >
```



6 8.Filter Doctor Name & Patient Satisfaction score using loc

[26]:
```python
df3 = df.loc[:,['Doctor_Name','Patient_Satisfaction_Score']]
df3
```

[26]:

	Doctor_Name	Patient_Satisfaction_Score
0	Dr. Mthembu	4
1	Dr. Naidoo	5
2	Dr. Khumalo	3
3	Dr. Daniels	4
4	Dr. Pillay	5
5	Dr. Mthembu	4
6	Dr. Naidoo	5

7	Dr. Khumalo	3
8	Dr. Daniels	4
9	Dr. Pillay	5
10	Dr. Mthembu	4
11	Dr. Naidoo	5
12	Dr. Khumalo	3
13	Dr. Daniels	4
14	Dr. Pillay	5
15	Dr. Mthembu	4
16	Dr. Naidoo	5
17	Dr. Khumalo	3
18	Dr. Daniels	4
19	Dr. Pillay	5
20	Dr. Mthembu	4
21	Dr. Naidoo	5
22	Dr. Khumalo	3
23	Dr. Daniels	4
24	Dr. Pillay	5
25	Dr. Mthembu	4
26	Dr. Naidoo	5
27	Dr. Khumalo	3
28	Dr. Daniels	4
29	Dr. Pillay	5
30	Dr. Mthembu	4
31	Dr. Naidoo	5
32	Dr. Khumalo	3
33	Dr. Daniels	4
34	Dr. Pillay	5
35	Dr. Mthembu	4
36	Dr. Naidoo	5
37	Dr. Khumalo	3
38	Dr. Daniels	4
39	Dr. Pillay	5
40	Dr. Mthembu	4
41	Dr. Naidoo	5
42	Dr. Khumalo	3
43	Dr. Daniels	4
44	Dr. Pillay	5
45	Dr. Mthembu	4
46	Dr. Naidoo	5
47	Dr. Khumalo	3
48	Dr. Daniels	4
49	Dr. Pillay	5

7 9. Doctor Performance Based on Patient Satisfaction Scores

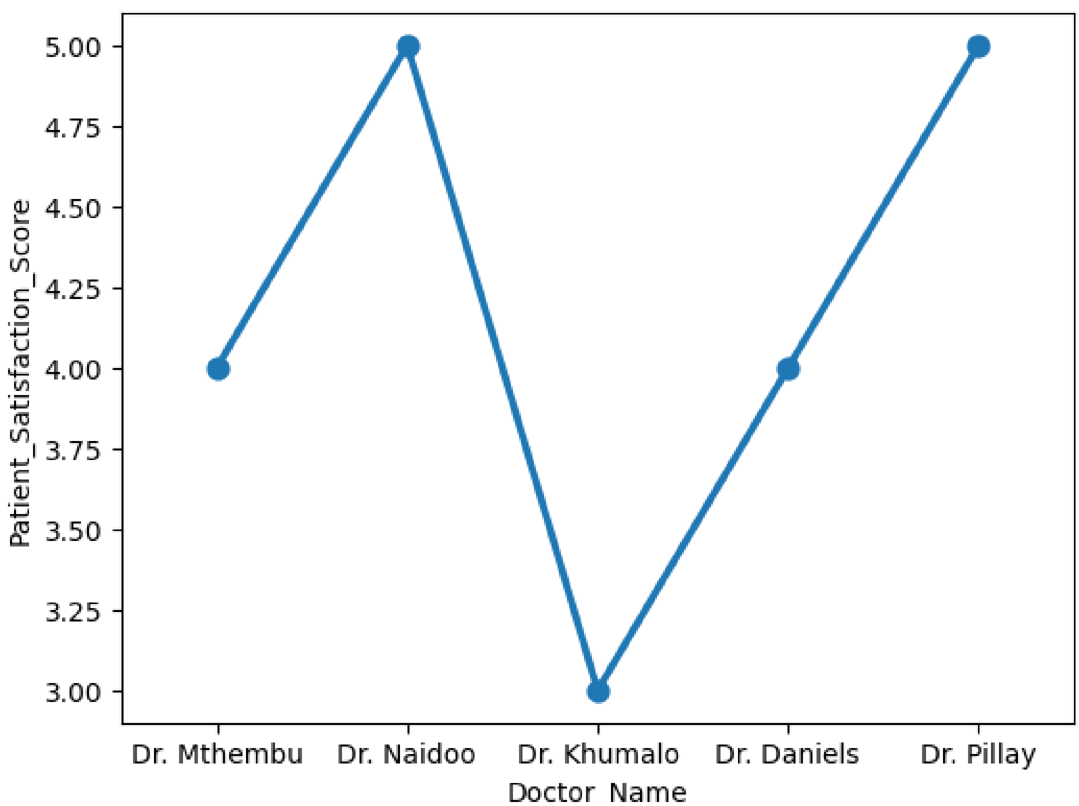
The point plot illustrates the average patient satisfaction scores for each doctor, highlighting noticeable differences in performance. Dr. Naidoo and Dr. Pillay stand out as the top performers, both achieving the highest average satisfaction score of 5. This indicates consistently excellent patient experiences and strong service delivery from these two doctors. Dr. Mthembu and Dr. Daniels demonstrate moderate performance, each with an average score of 4. Their scores suggest generally positive patient feedback, although there is room for improvement to reach top-tier performance levels. In contrast, Dr. Khumalo records the lowest average patient satisfaction score of 3, making this doctor the least performer among the group. This lower score signals potential challenges in patient experience and highlights an opportunity for targeted interventions, such as additional training or process improvements. Overall, the graph clearly distinguishes high, average, and low performers, making it a useful tool for management to recognize excellence, support improvement, and guide performance-enhancement strategies.

[30]:
```
df3.columns
```

[30]:
```
Index(['Doctor_Name', 'Patient_Satisfaction_Score'], dtype='str')
```

[42]:
```
sns.pointplot(df3, x= 'Doctor_Name', y= 'Patient_Satisfaction_Score')
```

[42]:
```
<Axes: xlabel='Doctor_Name', ylabel='Patient_Satisfaction_Score'>
```



8 10.Diagnosis vs Treatment Costs

[60]:
```
df.columns
```

[60]:
```
Index(['Patient_ID', 'Department', 'Visit_Date', 'Doctor_Name', 'Diagnosis',
       'Treatment_Cost', 'Patient_Satisfaction_Score'],
      dtype='str')
```

[75]:
```
df5 = df.loc[:,['Department', 'Treatment_Cost']]
df5
```

[75]:

	Department	Treatment_Cost
0	General Medicine	350
1	Pediatrics	200
2	Emergency	2200
3	Surgery	5000
4	Pharmacy	120
5	General Medicine	350
6	Pediatrics	200
7	Emergency	2200
8	Surgery	5000
9	Pharmacy	120
10	General Medicine	350
11	Pediatrics	200
12	Emergency	2200
13	Surgery	5000
14	Pharmacy	120
15	General Medicine	350
16	Pediatrics	200
17	Emergency	2200
18	Surgery	5000
19	Pharmacy	120
20	General Medicine	350
21	Pediatrics	200
22	Emergency	2200
23	Surgery	5000
24	Pharmacy	120
25	General Medicine	350
26	Pediatrics	200
27	Emergency	2200
28	Surgery	5000
29	Pharmacy	120
30	General Medicine	350
31	Pediatrics	200
32	Emergency	2200
33	Surgery	5000
34	Pharmacy	120

```
35   General Medicine             350
36         Pediatrics             200
37          Emergency            2200
38            Surgery            5000
39           Pharmacy             120
40   General Medicine             350
41         Pediatrics             200
42          Emergency            2200
43            Surgery            5000
44           Pharmacy             120
45   General Medicine             350
46         Pediatrics             200
47          Emergency            2200
48            Surgery            5000
49           Pharmacy             120
```

[77]:
```python
df6= df5.groupby('Department').sum()
df6
```

[77]:
```
                  Treatment_Cost
Department
Emergency                  22000
General Medicine            3500
Pediatrics                  2000
Pharmacy                    1200
Surgery                    50000
```

[86]:
```python
df5.columns
```

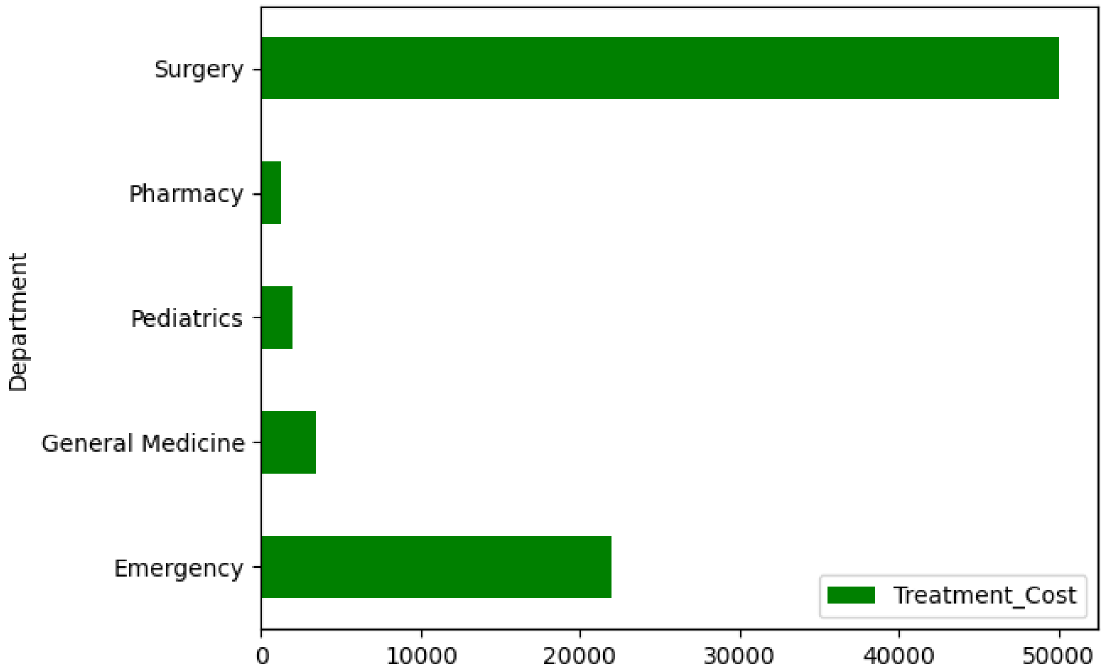
[86]:
```
Index(['Department', 'Treatment_Cost'], dtype='str')
```

9 Departmental Treatment Cost Analysis

The bar chart presents the total treatment cost incurred by each hospital department, revealing significant differences in resource utilization across departments. The Surgery department records by far the highest total treatment cost, amounting to 50,000. This indicates that surgical services are the most resource-intensive, likely due to complex procedures, specialized equipment, operating theatre costs, and extended post-operative care. As a result, surgery represents the hospital's largest cost driver. The Emergency department follows with a total cost of 22,000, making it the second-highest contributor to overall treatment expenses. This reflects the urgent and unpredictable nature of emergency care, which often requires immediate diagnostics, intensive treatment, and round-the-clock staffing. In contrast, General Medicine incurs a significantly lower cost of 3,500, suggesting that most treatments in this department are routine and less expensive compared to emergency or surgical care. Similarly, Pediatrics, with a total cost of 2,000, represents a relatively small share of overall expenditure, indicating efficient or less resource-heavy treatments for younger patients. The Pharmacy department has the lowest treatment cost, totaling 1,200, showing that medication costs alone contribute minimally compared to procedural and emergency-based services. Overall, the chart highlights that hospital costs are heavily concentrated in high-intensity departments, particularly Surgery and Emergency. This insight can support hospital management in budgeting, cost control, and strategic planning by focusing efficiency efforts on departments with the greatest financial impact.

[92]:
```
df6.plot.barh(color = 'green')
```

[92]:
```
<Axes: ylabel='Department'>
```



[ ]: